

Use of Air-Core Single-Phase Series Reactors to Reduce Medium-Voltage Switchgear Short-Circuit Duty

Technical White Paper

Prepared, reviewed, and issued by Consolidated Engineering and Sourcing Consultants (CESC)
Engineering note on the use of current-limiting reactors for MV fault-level mitigation

Scope	IEC-based engineering note covering technical principle, preliminary sizing, worked examples, and implementation checks for MV switchgear applications.
Primary standards basis	IEC 60909-0, IEC 60076-6, IEC 62271-200, with installation considerations linked to IEC 61936-1.
Primary conclusion	Air-core single-phase series reactors are a technically recognized way to reduce prospective short-circuit current and retain existing MV switchgear when full replacement is not economically justified.

Issued for technical guidance and engineering discussion.

1. Executive Abstract

This white paper sets out the engineering approach used by Consolidated Engineering and Sourcing Consultants (CESC) when evaluating the use of air-core single-phase series reactors to reduce prospective short-circuit duty on medium-voltage switchgear. The method is based on IEC short-circuit calculation practice, reactor application principles, and practical retrofit considerations. In projects where network reinforcement, added transformer capacity, new generation, or stronger utility contribution raises the bus fault level beyond the installed switchgear rating, series reactors can provide a technically sound alternative to full switchgear replacement, provided that voltage drop, protection coordination, breaker duty, installation clearances, and mechanical design are comprehensively verified.

2. Applicable Standards and Reference Framework

IEC 60909-0:2016 - Short-circuit currents in three-phase AC systems. Used as the principal method for calculating balanced and unbalanced fault levels and equivalent short-circuit impedance.

IEC TR 60909-4:2021 - Application guidance supporting IEC 60909-0 calculations and practical interpretation.

IEC 60076-6:2007 - Power transformers - Part 6: Reactors. Covers series reactors, including current-limiting reactors.

IEC 62271-200:2021 - AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV and up to and including 52 kV. Defines the switchgear performance envelope that must remain satisfied after network modification.

IEC 61936-1:2021 - Power installations exceeding 1 kV AC. Relevant for design, erection, clearances, and safe installation practice.

ANSI/IEEE C37 family - Useful parallel reference for breaker duty, interrupting capability, and North American practice when contract documents also reference ANSI/IEEE.

3. Engineering Basis and Fault-Limiting Principle

In accordance with IEC 60909 methodology, the initial symmetrical short-circuit current at an MV bus is inversely proportional to the equivalent short-circuit impedance seen from the fault location. CESC uses this principle as the basis for evaluating whether a series reactor can reduce the calculated duty to within the switchgear withstand and interrupting capability.

$$I_k = U / (\sqrt{3} \times Z_{eq})$$

Where I_k is the three-phase short-circuit current, U is the system line-to-line voltage, and Z_{eq} is the equivalent positive-sequence short-circuit impedance referred to the fault point.

If a reactor is inserted in series with the source path contributing to the fault, the new equivalent impedance becomes:

$$Z_{eq,new} = Z_{system} + Z_{reactor}$$

and therefore:

$$I_{k,new} = U / (\sqrt{3} \times (Z_{system} + Z_{reactor}))$$

Accordingly, the added reactance reduces the current contribution seen by the switchgear. This is the central engineering mechanism behind the use of current-limiting reactors.

4. Preliminary Engineering Sizing Method

For concept design and budgetary assessment, the additional reactor impedance may be estimated from the existing calculated short-circuit level and the target switchgear rating. This first-pass approach is useful for option screening only; final acceptance requires a full network model, equipment data validation, and coordinated electrical studies.

$$Z_{existing} = U / (\sqrt{3} \times I_{sc,existing})$$

$$Z_{target} = U / (\sqrt{3} \times I_{sc,target})$$

$$Z_{reactor} = Z_{target} - Z_{existing}$$

For an inductive reactor operated at frequency f :

$$X_L = 2 \times \pi \times f \times L$$

$$L = X_L / (2 \times \pi \times f)$$

This simplified method is suitable for preliminary reactor selection only. Final reactor rating and location should be validated using a detailed IEC 60909 model together with load-flow, protection, and equipment duty checks in recognized power system study software.

5. Worked Examples

Case	System Voltage	Existing 3ph Fault Level	Target Fault Level	Approx. Reactor X per Phase
Example 1	11 kV	31.5 kA	25 kA	0.052 ohm
Example 2	33 kV	40 kA	31.5 kA	0.125 ohm

Example 1 - 11 kV board reduction from 31.5 kA to 25 kA

$$Z_{existing} = 11,000 / (1.732 \times 31,500) = 0.202 \text{ ohm}$$

$$Z_{target} = 11,000 / (1.732 \times 25,000) = 0.254 \text{ ohm}$$

$$Z_{reactor} = 0.254 - 0.202 = 0.052 \text{ ohm per phase}$$

$$\text{At } 50 \text{ Hz, } L = 0.052 / (2 \times \pi \times 50) = 0.165 \text{ mH per phase}$$

Example 2 - 33 kV board reduction from 40 kA to 31.5 kA

$$Z_{existing} = 33,000 / (1.732 \times 40,000) = 0.476 \text{ ohm}$$

$$Z_{target} = 33,000 / (1.732 \times 31,500) = 0.605 \text{ ohm}$$

$$Z_{reactor} = 0.605 - 0.476 = 0.129 \text{ ohm per phase}$$

$$\text{At } 50 \text{ Hz, } L = 0.129 / (2 \times \pi \times 50) = 0.411 \text{ mH per phase}$$

6. Typical Application Arrangements in MV Systems

- Incoming incomer reactor: used when the principal fault contribution comes from the upstream utility or main transformer path.
- Bus-coupler reactor: used when parallel bus operation or tie connection causes the fault level to exceed switchgear capability.
- Generator or major motor source path reactor: used when internal source contribution materially raises the bus fault level.
- Selected feeder reactor: used when limiting contribution to a particular downstream board or process bus is more effective than acting at the main board.

7. Essential Design Checks Prior to Final Design Freeze

Check Item	Engineering Requirement
Short-circuit verification	Confirm maximum and minimum fault levels, contribution by each source, peak making current, and short-time withstand compatibility of the retained switchgear.
Steady-state voltage drop	Check operating voltage depression at full load and under motor starting or large process load changes.
Protection coordination	Review relay pickup values, grading margins, bus differential performance where fitted, and the impact of reduced fault magnitude on sensitivity and operating time.
Breaker duty and transients	Verify interrupting performance, recovery voltage implications, and any switching transient issues introduced by the added reactance.
Thermal and mechanical duty	Confirm continuous current rating, fault duty, conductor temperature rise, support strength, vibration, and electrodynamic forces.
Installation and layout	Confirm phase spacing, metallic clearance, magnetic field effects on nearby steelwork, support frames, foundations, and access for maintenance.
System operation	Review load sharing, bus transfer performance, generator synchronization implications, and any need for revised operating procedures.

8. Engineering Advantages and Practical Limitations

Advantages

- Can avoid full switchgear replacement where the exceedance above rating is moderate.
- Normally faster to implement than replacing a complete MV lineup and associated civil interfaces.
- Well suited to retrofit work where outage duration and plant disruption must be minimized.
- Technically straightforward to model under IEC 60909 using equivalent impedance methods.

Limitations

- Introduces additional voltage drop during normal operation.
- May influence protection sensitivity, relay grading, and breaker transient performance.
- Requires physical space and suitable support/foundation arrangements.

- Is not a universal solution where the fault-level exceedance is very large or where process voltage regulation is critical.

9. Recommended Engineering Workflow

- 1. Develop or update the IEC 60909 short-circuit model for the actual network configuration.**
- 2. Identify the exact source path causing the switchgear rating exceedance.**
- 3. Pre-size the reactor reactance from existing and target fault levels.**
- 4. Optimize the connection point to balance fault reduction against voltage drop and operational constraints.**
- 5. Re-run short-circuit, load flow, motor starting, and protection coordination studies.**
- 6. Confirm switchgear, breaker, protection, layout, structural, and installation compliance.**
- 7. Freeze the rating schedule and issue updated single line diagrams, protection settings, and operating notes.**

10. CESC Engineering Conclusion

From an engineering standpoint, air-core single-phase series reactors are a valid and well-established method for reducing prospective MV short-circuit current where the exceedance above switchgear rating is moderate and the network can tolerate the resulting added reactance. CESC would normally recommend this solution only after confirming, through formal system studies, that the revised arrangement satisfies fault-duty limits, voltage regulation, relay coordination, breaker performance, civil and layout constraints, and maintainability requirements. Where these checks are satisfactory, the reactor option can provide a practical and commercially efficient retrofit path while retaining the existing switchgear lineup.

11. Standards and Technical References

- IEC 60909-0:2016, Short-circuit currents in three-phase AC systems.
- IEC TR 60909-4:2021, guidance for application of IEC 60909-0.
- IEC 60076-6:2007, Power transformers - Part 6: Reactors.
- IEC 62271-200:2021, AC metal-enclosed switchgear and controlgear for rated voltages above 1 kV and up to and including 52 kV.
- IEC 61936-1:2021, Power installations exceeding 1 kV AC.
- Hitachi Energy, Series Reactors product and application overview, official technical product page.
- GE Vernova, Reducing Fault Currents in Power Systems, technical white paper.
- CIGRE/utility application literature on current-limiting reactors in feeder and bus-coupler applications.